# GSDDet: Ground Sample Distance-Guided Object Detection for Remote Sensing Images

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***Abstract*— Object detection for remote sensing images (ODRSI) is an important task in computer vision. Effective algorithms inspired by oriented object detection have been proposed recently. However, a major challenge still remains. Different categories of objects may be similar under different scales, causing cross-scale confusion. Different from natural images, remote sensing images have a consistent scale within the same image, which is generally referred to as ground sample distance (GSD). In this article, we show that GSD can be used to address the cross-scale confusion problem and effectively boost the performance of ODRSI. Specifically, we propose GSDDet, which embeds the deep features that represent GSD constraints to decrease the cross-scale confusion between different object categories. In GSDDet, a deep GSD classification network is first designed to extract the GSD deep features from remote sensing images. Then, the GSD deep feature is coupled with an attention framework to detect multiple categories of objects. Due to the simplicity of our framework, GSDDet can be applied to improve both the one-stage and two-stage methods. Experiments demonstrate that GSDDet outperforms state-of-the-art methods on challenging benchmarks, including DOTA-v1.0, DOTA-v1.5, and HRSC2016. The source code will be released upon publication.**

***Index Terms*— Remote sensing object detection.**

## I. Introduction

OBJECT detection for remote sensing images (ODRSI) plays an essential role in computer vision [1] and holds significant importance in remote sensing imagery analysis [2], [3], [4], [5]. In contrast to the majority of conventional natural images, remote sensing images are acquired from an elevated position, typically ranging between 300 and 1000 km above the Earth's surface. Consequently, the objects depicted in these images often possess relatively diminutive proportions, exhibit diverse rotations, and are encompassed by intricate backgrounds, thereby posing a formidable obstacle for object detection. The arduousness of this task from the elevated altitude and distinct angles at which the images are captured leads to perceptible disparities in the appearance of objects compared with conventional imagery. Therefore, detecting objects in remote sensing images requires specialized techniques that account for these distinctive complexities.

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To cope with these difficulties, deep learning methods have played a significant role in facilitating the swift advancement of remote sensing image interpretation, as a means of addressing the challenges associated with this field [6]. Most classical approaches [7], [8] primarily concentrate on accurately locating densely packed small target objects with varying rotational angles. In recent research, skillfully developed detectors for oriented objects have showcased encouraging outcomes when applied to challenging remote sensing image datasets [1], [2], [9]. These methodologies can be divided into two broad categories, namely, anchor-based and anchor-free. In the anchor-based methods, rotated RoI warping [7], [10] is used and one of the most commonly used methods is to extract the rotation-invariant features [11], whereas the anchor-free methods typically adopt direct angle-based orientation regression [12], [13]. Within the realm of these methodologies, anchor-based techniques typically entail more computationally demanding procedures in contrast to anchor-free methodologies but exhibit superior outcomes. As a result of ongoing methodological advancements, numerous anchor-free approaches [13], [14] have now achieved comparable levels of accuracy to anchor-based methods. From the perspective of methodology, the process of object detection can be done in two different ways: using one-stage methods [15], [16] and two-stage methods [17]. In the one-stage methods, the features of an image are extracted using a convolutional neural network, following which the network directly predicts the positioning and categorization of objects contained within the image. Conversely, the two-stage approaches involve the initial generation of region proposals, which are subsequently subjected to a convolutional neural network for the purpose of predicting the classification and positioning of objects within these proposals. Essentially, these two methods diverge in their fundamental approaches to addressing the task of object detection.

Nevertheless, the existing object detection methods for remote sensing adopt the conventional one-stage and two-stage approaches, failing to fully exploit the abundant geographical information embedded in remote sensing images. These approaches tend to disregard the inherent spatial context within the data, leading to confusion between different object categories due to the existence of comparable shapes or patterns across various scales. As shown in Fig. 1, the misidentification of roofs as entities like vehicles or other rectangular objects, along with the ambiguous recognition of certain circular objects, exemplify this problem. Therefore, there arises a necessity for the development of more sophisticated techniques capable of harnessing the spatial characteristics of remote

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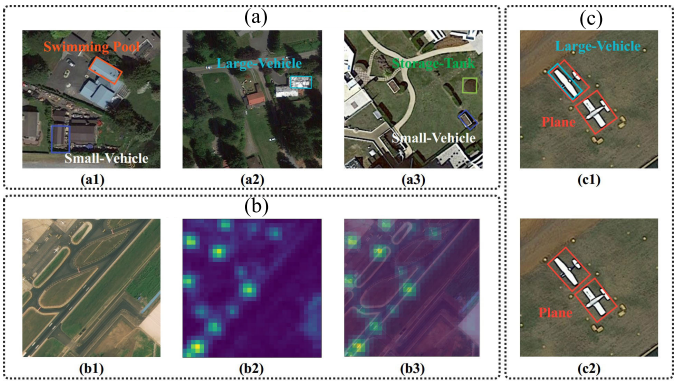



Fig. 1. **Illustration of the cross-scale confusion and heat map of GSD feature.** (a1)–(a3) Cross-scale confusion examples. (a1) Small vehicle was misidentified as the dark roof and the swimming pool was misidentified as the light roof. (a2) Large vehicle was misidentified as the white roof. (a3) Rectangular lawn was mistakenly identified as a small vehicle and a circular lawn was mistakenly identified as a storage tank. (b1) Remote sensing image. (b2) Heat map of deep ground sample distance (GSD) features extracted from the image. (b3) RoI for the GSD network when extracting features is clearly depicted.

sensing imagery in a highly efficient manner, to enhance the precision of object detection.

In contrast to natural images, remote sensing images exhibit a distinct characteristic wherein the pixels are uniformly represented by a standardized spatial resolution. This resolution, known as the ground sample distance (GSD), imposes a limitation on image clarity due to the inherent sampling process. The GSD precisely quantifies the physical distance between pixel centers on the ground, as observed through digital photographs of the Earth's surface acquired from either airborne or space-based platforms. To extend this idea, the ground sampling distance of a remote sensing image has a direct impact on the level of detail that can be discerned within the image. When the GSD value is smaller, the distance between pixel centers on the ground is reduced, resulting in the ability to capture a higher level of detail in the image. Conversely, larger GSD values correspond to a diminished capacity for capturing fine details. Therefore, when selecting the GSD for a remote sensing image, it is crucial to strike a balance between the necessity for detail and other considerations, such as storage and processing requirements. The discrimination of fluxes is of utmost importance for landscapes characterized by substantial variations in vegetation cover, architectural patterns, and moisture levels, making GSD a vital tool in this regard [18]. As stated by Hoffman and Markman [19], an initial crucial stage in the analysis of remote sensing images involves the identification of inherent data within the image, specifically encompassing spatial resolution. Previous studies [20], [21], [22], [23] also indicate that the comprehension of remote sensing images is influenced significantly by the spatial resolution, highlighting the fundamental role of GSD.

Inspired by the aforementioned observations, we introduce a novel object detection method named GSDDet, which uses the ground sample distance (GSD) as a guiding principle to enhance the accuracy of object detection in remote sensing images. This approach aims to address the problem of erroneous identification that emerges as a result of the confusion caused by varying scales within the image. GSDDet consists of two branches: the first branch is a deep GSD classification network, while the other branch is an object detector. The first branch uses a deep learning model to extract features associated with ground sampling distance from the remote sensing image. These features are subsequently incorporated into the second branch using a GSD-guided attention mechanism, aiming to enhance the precision of object detection. By incorporating GSD constraints, we expect GSDDet to improve the precision of object detection in remote sensing imagery. According to the illustration in Fig. 1, the process of feature extraction leads to the generation of a heat map associated with the GSD deep feature, which serves as robust evidence. The given statement elucidates the ability of the visualization to provide an accurate portrayal of the ground sample distance area, akin to human perception. Moreover, this visual representation facilitates comprehension and analysis of the extracted features in a manner that is more intuitive and conducive to scholarly investigation.

In summary, our main *contributions* are listed below.

1) We propose a novel approach for object detection in remote sensing images by incorporating ground sample distance (GSD) constraints encoding as deep features to compress the cross-scale confusion.
2) Due to the simplicity of our framework, the pretrained ground sample distance classification network (GSDN) can be seamlessly integrated into both the one-stage and two-stage object detection methods. To our best knowledge, it is the first pipeline to be stacked on any one-stage and two-stage modern detection framework flexibly.
3) Extensive experiments performed on the challenging datasets DOTA-v1.0, DOTA-v1.5, and HRSC2016 demonstrate the effectiveness of our GSDDet.

The rest of this article is organized as follows: Section II presents the related studies in ODRSIs. In Section III, we introduce the proposed method in detail. Moreover, extensive experiments and discussions are presented in Section IV. Finally, Section V provides the conclusions.

## II. Related Work

### A. Object Detection in Remote Sensing Images

There are two primary categories of convolutional neural-network-based techniques for detecting objects in images: anchor-based and anchor-free detectors. The anchor-based category can be further classified into two-stage methods [24] and one-stage methods [25]. The two-stage method involves first generating a set of predefined anchor boxes and then refining the proposals to accurately identify objects. On the other hand, the one-stage method uses a single CNN to simultaneously generate object proposals and classify them. In contrast, the anchor-free approach does not rely on predefined anchor boxes but instead directly predicts the location and size of objects within an image. These distinctions are essential to consider when choosing an appropriate object detection method for a given task. The anchor-based techniques are used to represent object detection as a task that involves classifying and regressing some proposal regions. In the field of object detection for remote sensing imagery, recent advancements have been achieved through the adoption of R-CNN frameworks [26], [27].The proposed methods [28], [29] use horizontal bounding boxes (HBBs) as regions of interest (RoIs) to extract features

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and identify categories. In contrast to conventional images, remote sensing images frequently contain rotating objects that cause misalignment between the bounding boxes and instances when horizontal RoIs [7] are used.

In contrast to most general object detectors [15], [24] that use HBBs, oriented object detectors use oriented bounding boxes (OBBs) that provide plentiful orientation information, which can avoid the problem of misalignment in a certain. It is crucial for detecting aerial objects with large aspect ratios, arbitrary orientations, and dense distributions. With the development and inspiration of classic object detection, many well-designed methods have been proposed for oriented object detection, accompanied by a promising performance on challenging datasets [2], [9]. To enhance the precision of regression on objects with orientation, some methods [10], [30] have used multiple rotated anchors with varying angles, scales, and aspect ratios, with the downside of increasing the computation complexity. Ding et al. [7] introduced the RoI Transformer, a detection model consisting of three stages, namely, the RRoI Leaner, RRoI wrapping, and feature extraction for classification and regression. The RRoI Leaner is responsible for learning the transformation from HRoI to RRoI, while the RRoI wrapping is in charge of extracting rotationally invariant features from RRoI. Yang et al. [28] developed the SCRDet, which incorporates a sophisticated sampling method and feature fusion network, as well as an intersection over union (IoU) loss, to address the challenge of boundary estimation in the context of OBB regression. Gliding Vertex [31] and RSDet [32] are both quadrilateral detectors, which limit the offset on the corresponding side of HBBs and order the sequence of quadrilateral corners. Although [32] solved the order of corner points to a certain extent, there still exist boundary problems. Yang and Yan [29] concluded that current popular regression-based angle prediction methods all have boundary problems to varying degrees. The aforementioned issue was resolved through a transformation of the angle regression problem into a classification problem, coupled with the imposition of restrictions on the scope of the predictions. ReDet [11] is a novel module called rotation-equivariant RoI that enables the extraction of rotation-invariant features from equivariant ones and this process is carried out by adapting to the orientation of the RoI.

According to the original article on anchor-free methods by YOLO [15], the task of object detection can be framed as a regression problem, where the goal is to predict the spatially separated bounding boxes and associated class probabilities for each object in an image. This approach eliminates the need for predefined anchor boxes, which are commonly used in other object detection methods to simplify the task of bounding box regression. The network directly predicts the bounding boxes' coordinates, sizes, and object scores, which indicate the likelihood of a box containing an object. This method has shown promising results in terms of accuracy and speed compared with the traditional anchor-based methods. The FSAF [33] aims to enhance object detection accuracy by constructing anchor-free branches at every level of the feature pyramid. This approach eliminates the need for anchor boxes that are typically used in object detection models to detect objects at different scales and aspect ratios. DRN [34] detects target objects through dynamic feature selection and refinement. Comparing with anchor-based approaches, anchor-free methods have a much higher detection speed. Since there is no need to preset the anchor, only the target center point, width, and height of the feature map at different scales need to be regressed, which greatly reduces the time and computation consumption.

Previous methods for object detection in remote sensing images have mainly relied on either anchor-based or anchor-free approaches, which have been adapted from conventional object detection methods. However, these methods tend to overlook the unique spatial resolution characteristics of remote sensing images. As a result, there is a need for novel approaches that take into account the distinct features of remote sensing data to improve the accuracy and efficiency of ODRSIs.

### *B. Ground Sample Distance*

Ground sample distance (GSD) is a fundamental concept in remote sensing that describes the minimum discernible size of an object or feature in an image. It is a measure of the spatial resolution of a remote sensing system and is determined by the size of the pixels that make up the image. In other words, it refers to the distance on the ground that each pixel in the image represents. The GSD of an image is affected by several factors, including the altitude and orientation of the sensor, the wavelength of the electromagnetic radiation used, and the size and quality of the sensor. In addition, the GSD is inversely proportional to the area of the image covered by each pixel. Therefore, images with smaller pixel sizes have a higher GSD and can distinguish smaller features on the ground. Currently, the GSD of remote sensing image is a very critical factor to consider when doing analyses of vegetation [35], [36]. A vegetation analysis can be conducted by analyzing different ground sample distance images, which are of great significance to regional ecological environments and cannot be replaced by ordinary image information. In addition, in areas such as traffic planning and vehicle navigation, remote sensing imagery can be used to extract road information [37], [38]. It is considered suitable to use images with diverse spatial resolutions based on the specific application scenarios. For instance, a 2-m resolution image could be used to track alterations in land cover and land use. In contrast, images with higher resolution, such as 0.8 m, are more appropriate for monitoring road networks within urban areas. Moreover, images with even higher resolutions, ranging from 0.5 to 0.3 m, can be leveraged to investigate intricate details, such as road boundaries, turning arrows, and zebra crossings. The utilization of such varying resolutions in imagery is vital in ensuring the accurate interpretation of remote sensing data, thereby enabling a more informed decision-making process in various fields of application, including urban planning and environmental management. Based on the aforementioned observations, we propose an innovative method that uses ground sampling distance (GSD) to do the object detection within remote sensing images. This approach aims to enhance the accuracy and effectiveness of identifying and locating objects or events within such imagery. Using GSD, we can obtain a more precise understanding of the spatial resolution of the images, allowing for improved detection and analysis of features within the imagery.

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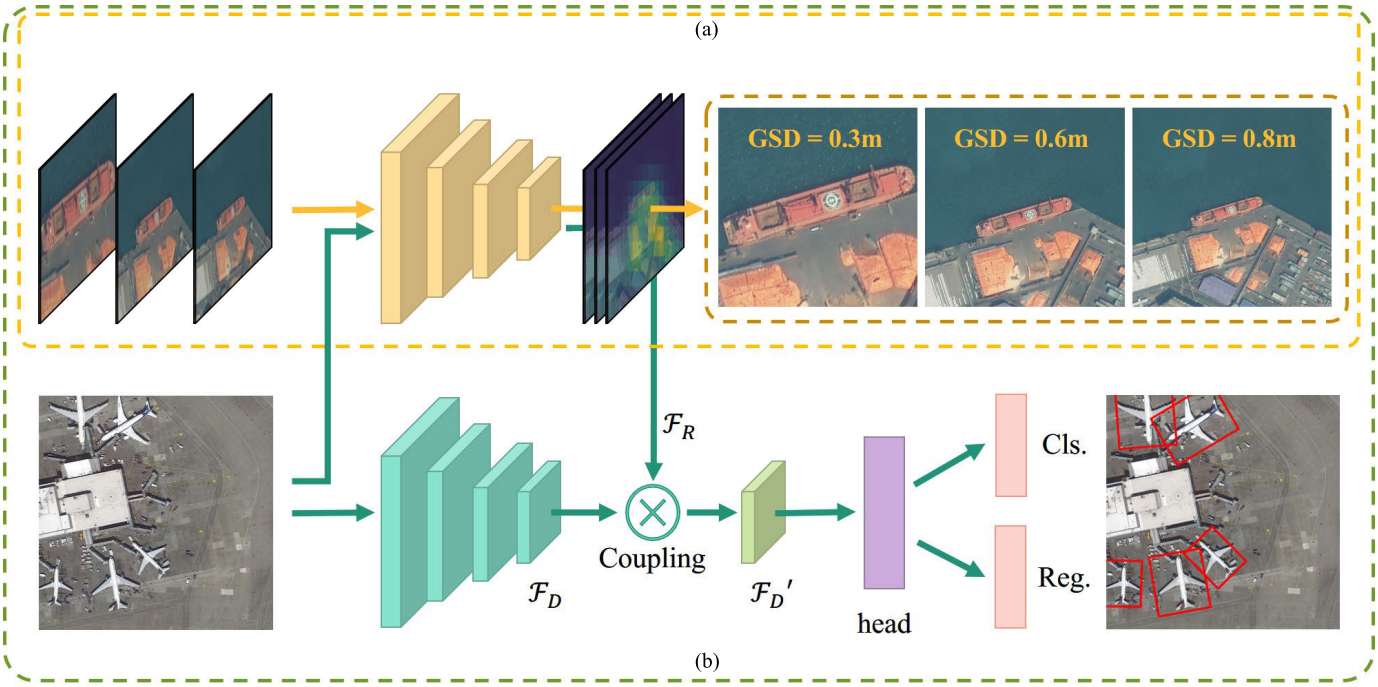



Fig. 2. **Framework of our proposed method. (a)** Ground sample distance classification network (GSDN). During the pretraining phase of GSDN, an image labeled with GSD is used as the input, and GSD is produced as the output. Once GSDN is pretrained, the classification head is removed and only the feature map $F_R$ generated by GSDN is used during the inference and subsequent training phases. **(b)** Proposed object detection branch. During both the inference and training of this branch, an image is first feed into both the branches to generate two feature maps, $F_D$ and $F_R$. Then we merge these two feature maps using a GSD-guided attention mechanism to produce a new feature map $F_D'$ and send it back to the detection head. *head* is a full-connected layer, *Cls.* is short for classification and *Reg.* is short for bounding box regression. Note that the dataset used to train the detection can be different from the one for pretraining GSDN, i.e., without GSD labels.

## III. Method

The GSDDet architecture, illustrated in Fig. 2, comprises two components: 1) a ***GSD*** *feature extraction* ***n****etwork* (*GSDN*) and 2) an *object detection network*. In the training phase, GSDN is first pretrained on datasets that contain GSD label, where the pretraining task involves classification. Specifically, the extracted GSD feature is fed into a multilayer perceptron [39] to predict the corresponding GSD value. Once the pretraining is completed, the MLP is discarded, and the GSDN weights are fixed, followed by training the object detection network on datasets that do not contain GSD label. During both inference and detection network training, each input image is processed by both the branches of GSDDet. The intermediate feature of the detection network is then combined with the GSD feature extracted from GSDN. The resulting merged feature is further used for classification and bounding box regression by the detection head. A GSD-guided attention mechanism is used to merge the features of two branches, as detailed in Section III-B. It is worth mentioning that GSDN is designed to extract GSD features that are more discriminative and informative for the subsequent object detection task. Pretraining GSDN on datasets with GSD label helps learn such features from the data that contain both GSD and object annotations. The pretraining phase can also improve the generalization ability of the GSDDet architecture, especially when the detection dataset has limited samples or high intraclass variance. In addition, by combining the intermediate feature of the detection network with the GSD feature, the GSDDet architecture can leverage the semantic information from GSDN to enhance object detection performance, particularly for objects that share similar appearance but belong to different GSD categories. The GSD-guided attention mechanism helps adaptively combine the features of two branches based on their relevance to GSD and object detection, respectively, which further improves the feature fusion process.

### A. GSD Feature Extraction Network (GSDN)

The first step of GSDN pretraining is to generate the labels on training data for GSD classification. As introduced in Section II, the ground sample distance of an image is calculated by

$$g = \frac{d_{\text{geo}}}{n} \tag{1}$$

where $g$ is the GSD, $d_{\text{geo}}$ is the actual distance on Earth, and $n$ is the number of pixels of the instance. Given a dataset with readable GSD information, we use the geospatial data abstraction library (GDAL) [40] to extract the GSD information for each image. The output of GDAL is a 6-D vector $\mathbf{v}$ for each image, where the second element in $\mathbf{v}$ is the corresponding GSD value $g$ of the image. $g$ is normally measured in meters. Moreover, when its value is extremely small, the obtained value from $\mathbf{v}$ can be measured in degrees, which we convert to meters $g$ via

$$g = \frac{2\pi r D}{360} \tag{2}$$

where $D$ is the GSD in degrees, and $r$ is the equatorial radius. We use $g$ as the classification labels for pretraining

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GSDN. Note that $g$ [0.3, 0.6, 0.8 m] in our training data (see Section IV-A for more details), and hence, pretraining is done via classification rather than regression.

In Fig. 2(a), the input is an image $\mathbf{I}$ and one of the output is a heat map of $\mathbf{I}$. For an input $\mathbf{I}$, the neural network predicts it to be a certain category. More details of the heat map are indicated in Fig. 3. Through the heat maps of the feature maps extracted by the GSD feature extraction network, some regions have a very visible color compared with the surrounding areas. Corresponding to the network, the more obvious the color of a region, the more significant its contributory value to the resolution feature extraction. Surprisingly, as shown in Fig. 3(a3), (b3), and (c3), most instances in these obvious regions have a high consistency with the categories labeled in the object detection task for remote sensing images. As a result of this correlation, we use GSD features to assist the task of ODRSIs.

Now we are ready to use the labeled data to pretrain GSDN. Denote $f(\cdot|\theta_G)$ as GSDN (without the classification head), and $h_C(\cdot|\theta_C)$ as the classification head (see Section IV-B for implementation details). Given an image $\mathbf{I}$, denote $\mathbf{o} = h_C(f(\mathbf{I}|\theta_G)|\theta_C)$ as the output of the classification head, the loss for pretraining is cross-entropy, i.e.,

$$l(\mathbf{o}, k_g) = -\log\left(\frac{e^{\mathbf{o}^{(k_g)}}}{\sum_j e^{\mathbf{o}^{(j)}}}\right) \tag{3}$$

where $k_g$ is the class index corresponding to the GSD label $g$.

After pretraining of GSDN, we drop the classification head ($h_C(\cdot|\theta_C)$) and fix the weights $\theta_G$ of GSDN during inference and training of the object detection branch. We do not further use the GSD label during training of the detection branch since the GSD value is not available in many remote sensing datasets. This allows our method to be applied generally to datasets that are different from the one that GSDN is pretrained on (see Section IV for more details).

Fig. 3 shows the heat map of the GSD features during GSD classification. The presented data comprise three distinct sets of images representing ground sampling distance (GSD) features. These sets are categorized as follows: Fig. 3(a1), (b1), and (c1) depicting the input images, (a2), (b2), and (c2) exhibiting the heat maps of GSD features, and (a3), (b3), and (c3) representing the superimposed images. The images are arranged in three separate rows, each emphasizing unique GSD features. The initial row is conspicuous for its depiction of prominent features such as swimming pools and roads. The second row primarily highlights harbors, while the third row accentuates small vehicles and roads. The areas that exhibit lighter shades are of greater significance when it comes to ground sampling distance (GSD) classification. It is noteworthy that a majority of these highlighted regions coincide with the object of interest, thereby validating the empirical evidence of GSDN's efficacy in enhancing detection performance.

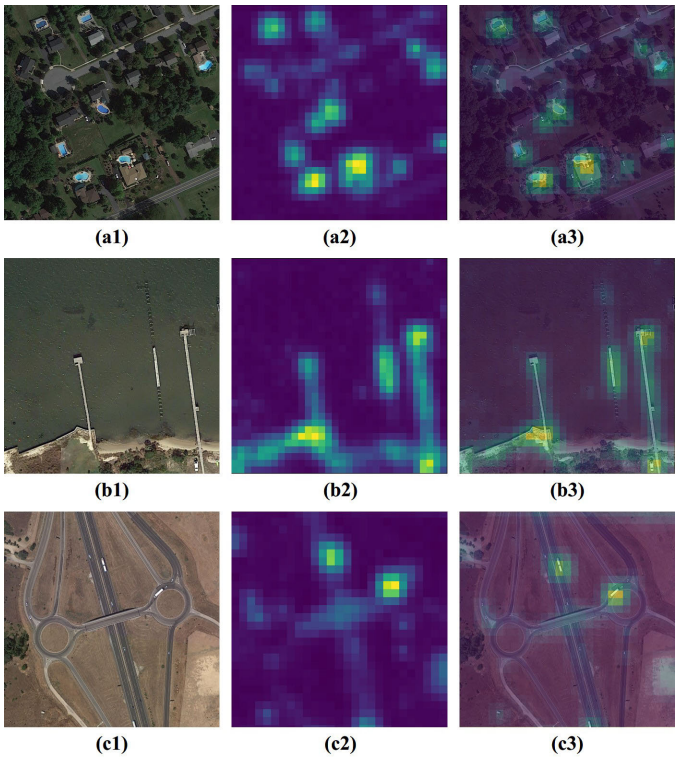



Fig. 3. **Heat maps of GSD features.** The given visual data comprise three sets of images denoting ground sampling distance (GSD) features, namely, (a1), (b1), and (c1) as input images, (a2), (b2), and (c2) as the corresponding heat maps of GSD features, and (a3), (b3), and (c3) as superimposed images. The images are arranged in three rows, each highlighting different features. The first row depicts the most apparent features such as swimming pools and roads. The second row showcases harbors as the primary feature, whereas the third row emphasizes small vehicles and roads.

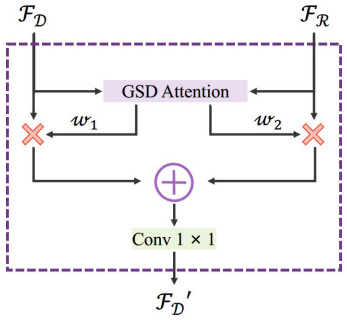



Fig. 4. **GSD-guided attention mechanism.** Instead of naively concatenating $\mathbf{F}_D$ and $\mathbf{F}_R$, we use (4) to effectively merge the GSD information with the detection branch.

## B. GSD-Guided Object Detection

As shown in Fig. 2(b), before feeding $\mathbf{F}_D$ into the detection head, we first merge it with the feature map $\mathbf{F}_R$ generated by GSDN, and then feed the merged feature $\mathbf{F}'_D$ into the final detection head. The loss for training the object detection network is the same as the one used to train a normal detection network.

Considering that the two branches of GSDDet have different structures and can be trained on different datasets (e.g., the object detection branch can be trained on datasets without GSD information), their corresponding feature maps can also have different dimensions. To make feature merging smooth for different detection datasets and baseline architectures, we adjust the GSDN architecture (without changing the fixed weights) before training the detection branch so that $\mathbf{F}_R$ and $\mathbf{F}_D$ have the same dimension. Specifically, assume the original output of GSDN is a feature map $\mathbf{F}_O \in \mathcal{R}^{D_1 \times D_2 \times D_3}$, and $\mathbf{F}_D \in \mathcal{R}^{D'_1 \times D'_2 \times D'_3}$, we first adjust the stride of the last one

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to two convolution layers of GSDN (without changing the weights) so that the first two dimensions of GSDN change from $D_1 \times D_2$ to $D_1' \times D_2'$. Then we add another convolution layer with $1 \times 1$ kernel size to GSDN to adjust the third dimension from $D_3$ to $D_3'$. This additional convolution layer is trained along with the detection branch.

We also propose a GSD-guided attention layer in the detection branch for feature merging, which in practice is more effective than naively concatenate two feature maps (see Section IV for detailed ablations). As shown in Fig. 4, we first compute the corresponding attention weights $\mathbf{w}_1$ and $\mathbf{w}_2$ on $\mathbf{F}_D$ and $\mathbf{F}_R$. Then, we generate the merged feature map $\mathbf{F}'_D$ via

$$\mathbf{F}'_D = \text{Conv}(\text{concat}(\mathbf{w}_1 \cdot \mathbf{F}_R, \mathbf{w}_2 \cdot \mathbf{F}_D)) \tag{4}$$

where

$$\mathbf{w}_1 = \text{Sigmoid}\big(\text{Att}_l(\mathbf{F}_R) \cdot \text{Att}_g(\mathbf{F}_R)\big) \tag{5}$$

$$\mathbf{w}_2 = \text{Sigmoid}\big(\text{Att}_l(\mathbf{F}_D) \cdot \text{Att}_g(\mathbf{F}_D)\big). \tag{6}$$

Conv is a $1 \times 1$ convolution to reduce the channel dimension. $\text{Att}_l$ and $\text{Att}_g$ are, respectively, the local and global attention [41]. Sigmoid is the sigmoid activation function.

Since the two GSDDet branches only interact at the feature map generation stage, our GSDDet framework can be effectively applied to both one-stage and two-stage methods, providing consistent improvements on different methods and datasets (see Section IV for demonstrations).

## IV. Experiments and Results

### A. Datasets

In our experiment, we use FAIR1M [1] for GSDN pretraining due to the rich geographical information available. We do not assume that the actual data for detection also contain the GSD info, and hence, we train the detection branch and evaluate different methods on other datasets, including different versions of DOTA [2] and HRSC2016 [9]. We provide the detailed introduction of each dataset below.

*1) FAIR1M:* FAIR1M is a large-scale dataset designed to refine object detection and recognition in remote sensing images. The images in FAIR1M were collected from Gaofen satellites and Google Earth with a ground sample distance range of 0.3–0.8 m. FAIR1M contains more than a million instances and more than 15 000 images. Five categories were annotated: airplanes, ships, vehicles, courts, and roads, with 37 subcategories within each category. There are ten types for airplanes, eight specific categories for ships, nine specific categories of vehicles, four categories for courts, three categories for roads, and "other-airplane," "other-ship," and "other-vehicle" for objects that do not belong to the previously defined specific object types. For airplanes, there are ten types including Boeing 737, Boeing 777, Boeing 747, Boeing 787, Airbus A320, Airbus A220, Airbus A330, Airbus A350, COMAC C919, and COMAC ARJ21. There are eight specific categories for ships, including passenger ships, motorboats, fishing boats, tugboats, engineering ships, liquid cargo ships, dry cargo ships, and warships. There are nine specific categories of vehicles, including small cars, buses, cargotrucks, dump trucks, vans, trailers, tractors, truck tractors, and excavators. Furthermore, there are four categories and three categories for courts and roads, respectively, and "other-airplane," "other-ship," and "other-vehicle" for objects that do not belong to the previously defined specific object types. Images range in size from $1000 \times 1000$ to $10\,000 \times 10\,000$ and contain objects of various sizes, orientations, and shapes.

*2) DOTA:* DOTA is a large-scale dataset for evaluating the detection performance of oriented object detection in aerial images with three released versions: DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0. DOTA-v1.0 consists of 2806 images, 188 282 instances, and 15 categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). It contains images of varying orientations, scales, and shapes. The size of these aerial images ranges from $800 \times 800$ to $4000 \times 4000$. There are 1411 images in the training set, 458 images in the validation set, and 937 images in the testing set. DOTA-v1.5 contains 402 089 instances with a new category, container crane (CC), released in 2019. The challenge of DOTA-v1.5 is greater than that of DOTA-v1.0. There are 1 793 658 instances and two new categories of airport (Air) and helipad (Heli) in DOTA-v2.0 with to DOTA-v1.5.

In our experiment, training sets and validation sets are used for training, and testing sets are used for testing. At the preprocessing stage, the original images are cropped into patches of $1024 \times 1024$ with an 824 stride. During the training stage, random resizing and flipping are performed to prevent overfitting.

*3) HRSC2016:* A large number of strip-like oriented objects have been collected for ship recognition from several famous harbors in HRSC2016 [9]. It contains 1061 aerial images with the size ranging from $300 \times 300$ to $1500 \times 900$. Our experiment uses a training set (436 images) and validation set (181 images) for training, and a testing set (444 images) for evaluation.

### B. Implementation Details

*1) GSDN:* Though more advanced architectures can also be applied, we choose ResNet50 (excluding the fully connected layer) as the architecture for GSDN, with a CBAM [48] layer (with channel attention and spatial attention) after the final convolution layer to promote the classification performance. The classification head for pretraining GSDN is a fully connected layer. We use the pretrained models from PyTorch [49] to initialize the GSDN weights and regulate the stride of the last layer according to the image size during GSDN pretraining. The test results are shown in Fig. 6. Among them, the accuracy of GSD = 0.3 and 0.8 m is much higher, but the accuracy of GSD = 0.6 m is lower due to the lack of experimental data. After data enhancement, the accuracy of GSD = 0.6 m is increased from 60% to 80%. We use an initial learning rate of 0.01 and a step learning rate decay schedule to modify $lr \leftarrow 0.1 \cdot lr$ every 50 epochs. The neural network architecture has been trained on the FAIR1M dataset for a duration of 100 epochs, USING a batch size of 32. The optimization algorithm used is stochastic gradient descent (SGD), where a momentum value of 0.9 has been set

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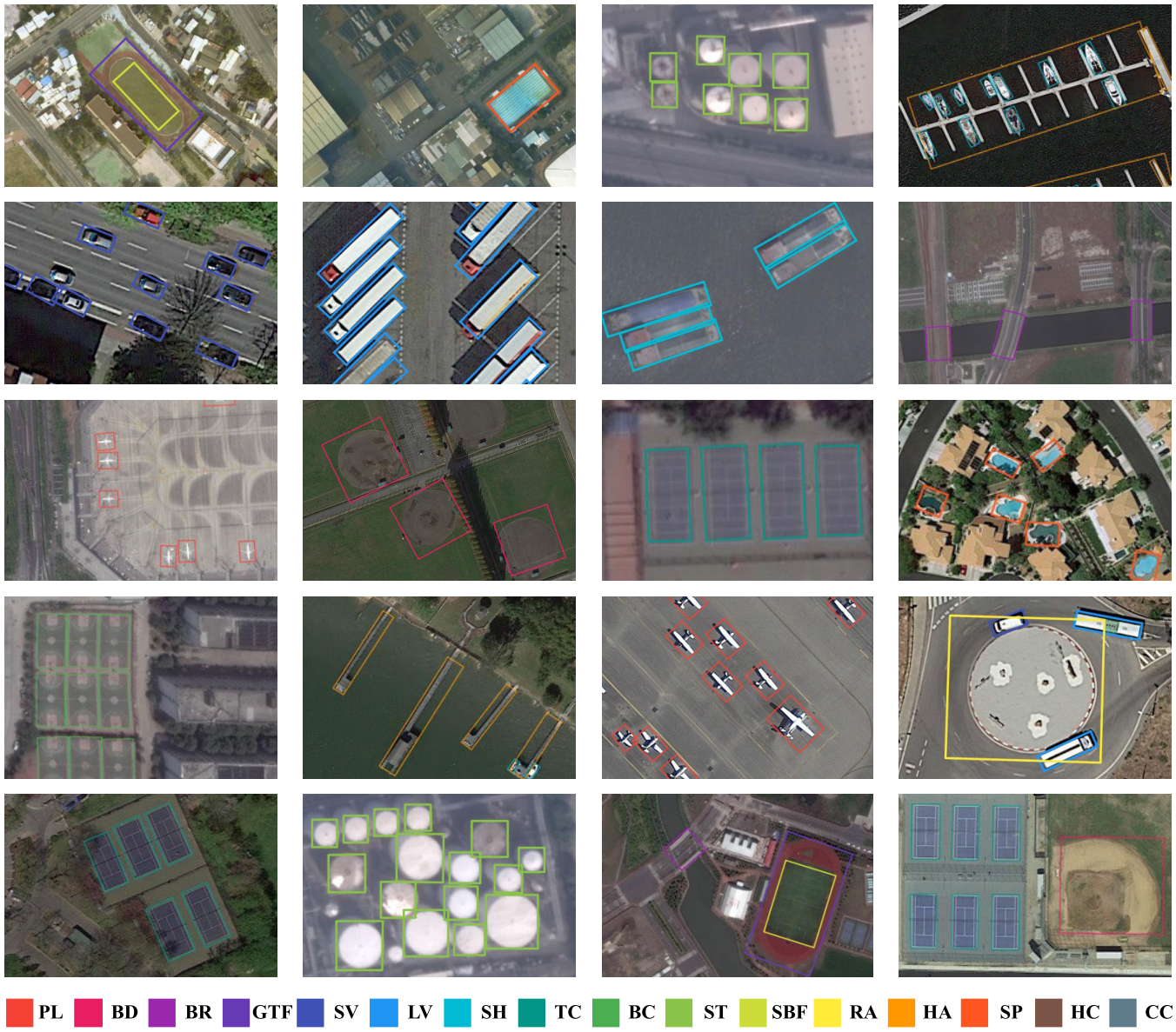



Fig. 5. Visualization results on single-scale DOTA-v1.5 with our GSDDet.

to improve convergence. The training and inference procedures are performed on a single V100 GPU.

*2) Object Detection:* To verify the universality of our method, we apply the GSDDet framework to two baselines, i.e., ReDet [11] and YOLO-v5x [50], where ReDet is two-stage and YOLO-v5x is one-stage. In this study, we adhere to the default hyperparameters for training the detection branch and testing for each method, unless specified otherwise. The training process uses four V100 GPUs, while a single V100 GPU is used for testing. The model is trained using a batch size of 64, 64, and 32 for DOTA-v1.0, DOTA-v1.5, and HRSC2016, respectively. The use of multiple GPUs for training allows for efficient processing of large datasets, while testing is performed on a single GPU for consistency and accuracy of results. The experiments are conducted on a server with a single V100 GPU to ensure fair comparison between different methods and to eliminate potential variations arising from the use of different hardware configurations.

### C. Ablation Studies

To examine the effectiveness of each component in our proposed framework, a series of ablation experiments are performed on DOTA-v1.0, DOTA-v1.5, and HRSC2016.

*1) Effectiveness of ground sample distance feature:* We introduce GSD features directly on both two-stage (ReDet) and one-stage (YOLO-v5x) detectors to evaluate the effectiveness of GSD features. To remove the effect of other components, e.g., the GSD-guided attention mechanism, we merge the features of the two branches by simple concatenation with a follow-up convolution layer. As illustrated in Tables I and II, naively incorporating deep GSD features already improved the performance of all the baselines by a considerable amount. With the help of GSD features, false detection by cross-scale confusion and mAP increased by 2.43% and 0.3%, respectively, in ReDet and YOLO-v5x. These improvements owe to the GSD constraint. Compared with ReDet, the complexity of

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TABLE I

COMPARISONS ON SINGLE-SCALE DOTA-V1.5 BETWEEN REDET†, ONLY GSD, AND OUR METHOD. † INDICATES THE MODEL IS OUR REIMPLEMENTED VERSION. NAIVELY CONCATENATING THE FEATURES FROM TWO BRANCHES (THE SECOND ROW) ALREADY IMPROVED THE PERFORMANCE OF THE BASELINE. WITH OUR GSD-GUIDED MECHANISM (THE THIRD ROW) THE PERFORMANCE IS FURTHER IMPROVED

| Method | GSD | Att. | mAP |
|---|---|---|---|
| ReDet† | | | 66.24 |
| +GSD | ✓ | | 68.67 (+2.43) |
| GSDDet (Ours) | ✓ | ✓ | 69.13 (+0.46) |

TABLE II

COMPARISONS ON SINGLE-SCALE DOTA-V1.5 BETWEEN YOLO-V5X†, ONLY GSD, AND OUR METHOD. NAIVELY CONCATENATING THE FEATURES FROM TWO BRANCHES (THE SECOND ROW) ALREADY IMPROVED THE PERFORMANCE OF THE BASELINE. WITH OUR GSD-GUIDED MECHANISM (THE THIRD ROW) THE PERFORMANCE IS FURTHER IMPROVED

| Method | GSD | Att. | mAP |
|---|---|---|---|
| YOLO-v5x† | | | 44.92 |
| +GSD | ✓ | | 45.22 (+0.30) |
| GSDDet (Ours) | ✓ | ✓ | 46.38 (+1.16) |

TABLE III

ABLATION STUDIES ON ONLY GSD AND UNDER GSD-GUIDED ATTENTION MECHANISM ON SINGLE-SCALE DOTA-V1.5. CERTAIN CATEGORIES WERE IDENTIFIED TO HAVE SUBOPTIMAL PERFORMANCE WHEN SOLELY USING THE GSD FEATURE, BUT NOTABLE ENHANCEMENT WAS ACHIEVED THROUGH THE INCORPORATION OF GSD-GUIDED ATTENTION MECHANISM

BD, GTF, SV and SBF denote the categories of Baseball Diamond, Ground Track Field, Small Vehicle and Soccer Ball Field, respectively.

| GSD | Att. | BD | GTF | SV | SBF |
|---|---|---|---|---|---|
| ✓ | | 80.40 | 69.47 | 51.49 | 60.92 |
| ✓ | ✓ | 82.93 | 74.18 | 52.33 | 63.50 |
| improvement | | +2.53 | +4.71 | +0.84 | +2.58 |

the YOLO-v5x structure and the reuse of detected features make the performance of improvement less obvious.

*2) Effectiveness of GSD-Guided Attention Mechanism:* The proposed approach incorporates a GSD-guided attention mechanism to prevent feature drowning during feature fusion. The experimental results in Tables I and II demonstrate that the performance has significantly improved compared with the conventional feature fusion method. Specifically, the YOLO-v5x model has shown the most improvement. In addition, Table III shows that the inclusion of this mechanism has resulted in substantial enhancements for several categories in DOTA-v1.5, including baseball diamond, ground track field, small vehicle, and soccer-ball field. These improvements indicate the efficacy of the GSD-guided attention mechanism in enhancing the accuracy of object detection in aerial images.

*3) Performance on Different Datasets:* We evaluate the performance of reimplemented version ReDet on single-scale DOTA-v1.0 and DOTA-v1.5, and reimplemented version YOLO-v5x on single-scale DOTA-v1.5 to demonstrate generalization. Some of the visualisation results are shown in Fig. 5. For YOLO-v5x, we train for 45 epochs. As shown

TABLE IV

PERFORMANCE ON SINGLE-SCALE DOTA-V1.5 IN COCO STYLE. YOLO-V5X† IS OUR REIMPLEMENTED VERSION

| Method | DOTA-v1.5 | | |
|---|---|---|---|
| | AP50 | AP75 | mAP |
| YOLO-v5x† | 44.91 | 28.00 | 28.04 |
| Ours | 46.37 | 28.71 | 28.96 (+0.92) |

TABLE V

PERFORMANCE OF THE PROPOSED GSDDET ON OTHER DATASETS. WE REPORT THE PERFORMANCE ON SINGLE-SCALE DOTA-V1.0 AND SINGLE-SCALE DOTA-V1.5 IN COCO STYLE. REDET† IS OUR REIMPLEMENTED VERSION

| Method | DOTA-v1.0 | | | DOTA-v1.5 | | |
|---|---|---|---|---|---|---|
| | AP50 | AP75 | mAP | AP50 | AP75 | mAP |
| ReDet† | 75.43 | 50.13 | 46.70 | 66.24 | 41.01 | 40.01 |
| Ours | 76.57 | 51.77 | 47.52 (+0.82) | 69.13 | 42.96 | 41.48 (+1.47) |

in Tables IV and V, compared with the baselines, our method achieves better performance on both the datasets. Furthermore, improvements have been made in *AP75* and *mAP*, which demonstrate its accuracy in localization.

### D. Comparison

*1) Results on DOTA-v1.0:* As shown in Table VI, we conducted a comprehensive evaluation of the GSDDet method on the DOTA-v1.0 OBB task, comparing its performance against other state-of-the-art methods. We used both single-scale and multiscale models and evaluated their respective performance. Our single-scale model achieved a mean average precision of **76.57**, which surpassed the performance of all other single-scale models and the majority of multiscale models. Moreover, we implemented limited data augmentations to further enhance the performance of our multiscale model, resulting in a state-of-the-art performance of **80.45** mAP. Notably, our approach yielded particularly positive results for the **GTF** and **SBF**.

*2) Results on DOTA-v1.5:* Compared with DOTA-v1.0, DOTA-v1.5 contains more tiny instances, making object detection more challenging. Our method outperformed other single-scale methods by a wide margin in single-scale data, achieving **69.13** mAP. As shown in Table VII, we obtained the best results among **9/16** categories, and the best or second-best results among **13/16** categories. When compared with ReDet in single-scale, GTF increased by **2.77%**, SH increased by **6.24%**, BC increased by **8.33%**, SBF increased by **14.21%**, and CC increased by **3.62%**. The visualization comparison between ReDet and our method is shown in Fig. 7. Using GSDDet, we were able to detect objects that have not been correctly detected before. Specifically, we observed notable improvements in detecting smaller vehicles, tennis courts, harbors, and bridges. In addition, we noted a reduction in errors resulting from cross-scale fusion, particularly between dissimilar categories such as large vehicles and planes, planes and helicopters, swimming pools and basketball courts, and baseball diamonds and roundabouts. Overall, our experimental results demonstrate the efficacy of the GSDDet approach in object detection and highlight its potential to enhance the accuracy and reliability of object detection systems.

In comparison to previous best results, our multiscale model has demonstrated exceptional performance, reaching

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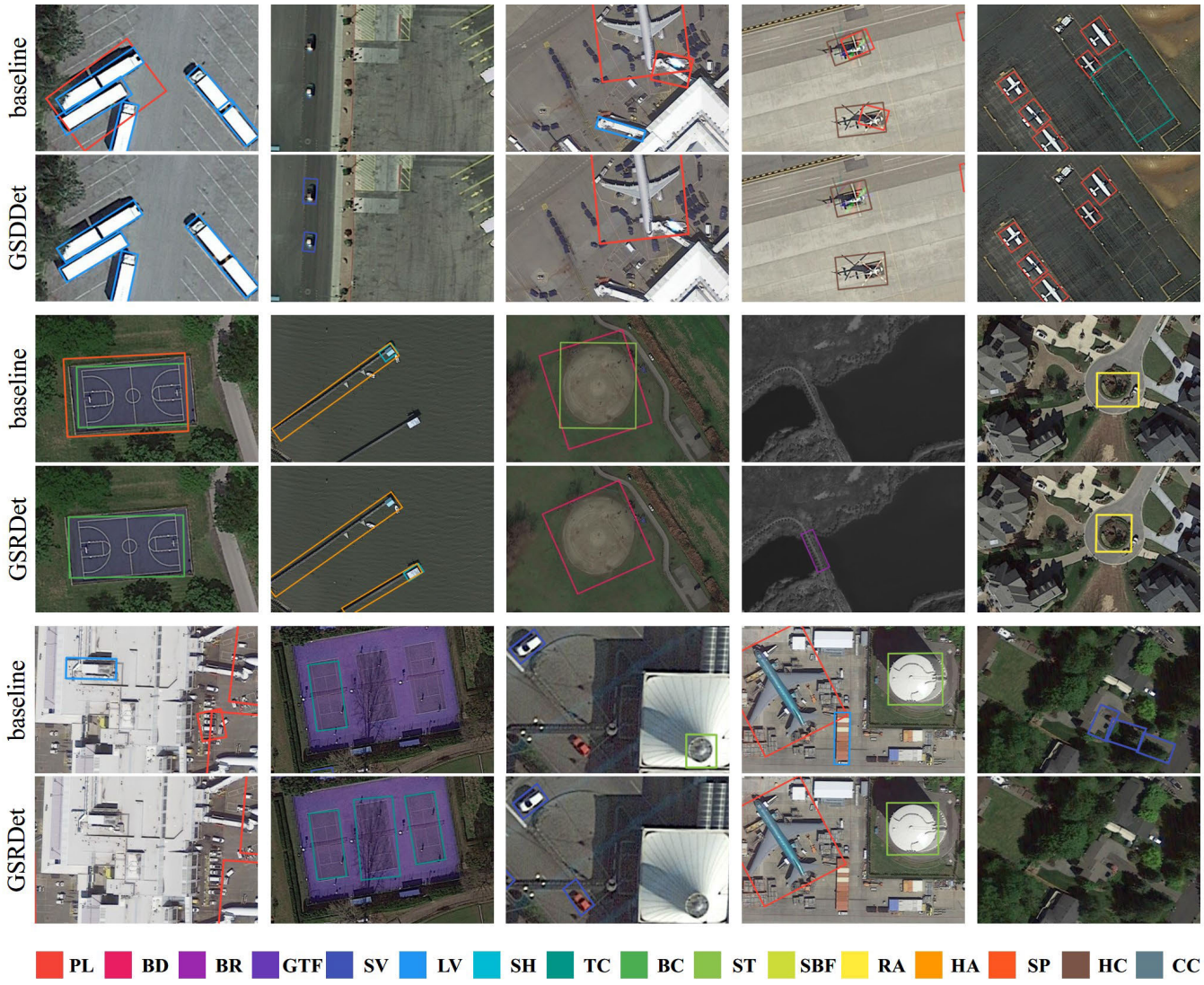



Fig. 6. **Comparison of visualization results between ReDet and our GSDDet on single-scale DOTA-v1.5.** By incorporating the explicitly encoded GSD feature, GSDDet exhibited fewer errors resulting from cross-scale confusion. The variously colored boxes depicted in the illustration correspond to distinct categories enumerated at the bottom.

TABLE VI

COMPARISONS WITH STATE-OF-THE-ART METHODS ON DOTA-V1.0 OBB TASK. ∗ INDICATES MULTISCALE TRAINING AND TESTING. BOLD FONT FOR THE BEST RESULT AND UNDERLINE FOR THE SECOND-BEST RESULT

| Methods | PL | BD | BR | GTF | SV | LV | SH | TC | BC | ST | SBF | RA | HA | SP | HC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **single-scale:** | | | | | | | | | | | | | | | | |
| GSDet [20] | 81.12 | 76.78 | 40.78 | **75.89** | 64.50 | 58.37 | 74.21 | 89.92 | 79.40 | 78.83 | 64.54 | 63.67 | 66.04 | 58.01 | 52.13 | 68.28 |
| RoI-Trans. [7] | 88.65 | 82.60 | 52.53 | 70.87 | 77.93 | 76.67 | 86.87 | 90.71 | 83.83 | 82.51 | 53.95 | **67.61** | 74.67 | 68.75 | 61.03 | 74.61 |
| DRN [34] | **88.91** | 80.22 | 43.52 | 63.35 | 73.48 | 70.69 | 84.94 | 90.14 | 83.85 | 84.11 | 50.12 | 58.41 | 67.62 | 68.60 | 52.50 | 70.70 |
| CenterMap-Net [5] | 88.88 | 81.24 | 53.15 | 60.65 | **78.62** | 66.55 | 78.10 | 88.83 | 77.80 | 83.61 | 49.36 | 66.19 | 72.10 | **72.36** | 58.70 | 71.74 |
| SCRDet [28] | 89.49 | 81.17 | 50.53 | 66.10 | 70.92 | 78.66 | 78.21 | 90.81 | 85.26 | 84.23 | 61.81 | 63.77 | 68.16 | 69.83 | 67.17 | 73.74 |
| RSDet++ [42] | 86.8 | 82.7 | **54.6** | 71.7 | 76.0 | 71.2 | 83.5 | 87.4 | 83.4 | 85.3 | **72.4** | 62.9 | 70.9 | 72.3 | **70.4** | 75.4 |
| ReDet [11] | 88.79 | 82.64 | 53.97 | 74.00 | 78.13 | **84.06** | **88.04** | **90.89** | **87.78** | **85.75** | 61.76 | 60.39 | **75.96** | 68.07 | 63.59 | 76.25 |
| GSDDet (Ours) | 88.83 | **83.34** | 53.39 | 75.24 | 77.83 | 83.03 | **88.04** | 90.86 | 87.47 | 85.73 | 65.25 | 64.17 | 75.56 | 67.62 | 62.24 | **76.57** |
| **multi-scale:** | | | | | | | | | | | | | | | | |
| DRN* [34] | 89.71 | 82.34 | 47.22 | 64.10 | 76.22 | 74.43 | 85.84 | 90.57 | 86.18 | 84.89 | 57.65 | 61.93 | 69.30 | 69.63 | 58.48 | 73.23 |
| Mask OBB* [43] | 89.61 | 85.09 | 51.85 | 72.90 | 75.28 | 73.23 | 85.57 | 90.37 | 82.08 | 85.05 | 55.73 | 68.39 | 71.61 | 69.87 | 66.33 | 74.86 |
| Gliding Vertex* [31] | 89.64 | 85.00 | 52.26 | 77.34 | 73.01 | 73.14 | 86.82 | 90.74 | 79.02 | 86.81 | 59.55 | **70.91** | 72.94 | 70.68 | 57.32 | 75.02 |
| Oriented R-CNN* [17] | 89.46 | 82.18 | 54.78 | 70.86 | 78.93 | 83.00 | 88.20 | **90.90** | 87.50 | 84.68 | 63.97 | 67.69 | 74.94 | 68.84 | 52.28 | 75.87 |
| CSL* [29] | **90.25** | **85.53** | 54.64 | 75.31 | 70.44 | 73.51 | 77.62 | 90.84 | 86.15 | 86.69 | 69.60 | 68.04 | 73.83 | 71.10 | 68.93 | 76.17 |
| ReDet* [11] | 88.81 | 82.48 | **60.83** | 80.82 | 78.34 | 86.06 | 88.31 | 90.87 | **88.77** | 87.03 | 68.65 | 66.90 | **79.26** | 79.71 | 74.67 | 80.10 |
| GWD* [44] | 89.66 | 84.99 | 59.26 | **82.19** | **78.97** | 84.83 | 87.70 | 90.21 | 86.54 | 86.85 | **73.47** | 67.77 | 76.92 | 79.22 | **74.92** | 80.23 |
| GSDDet* (Ours) | 88.95 | 84.78 | 60.69 | 80.01 | 78.38 | **86.17** | **88.37** | **90.90** | 88.12 | **87.18** | 72.99 | 67.78 | 79.06 | **80.54** | 72.77 | **80.45** |

a state-of-the-art mean average precision of **77.68**. Notably, specific categories within the model displayed remarkably high performance results. Moreover, we have included the CC category, recognized as the most challenging, in DOTA-v1.5.

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TABLE VII

COMPARISONS WITH STATE-OF-THE-ART METHODS ON DOTA-V1.5 OBB TASK. RETINANET-O, FR-O, AND MASK R-CNN ARE REIMPLEMENTED VERSION FOR DOTA-V1.5 BY HAN ET AL. [11]. OWSR IS A METHOD FROM DOAI 2019. ∗ INDICATES MULTISCALE TRAINING AND TESTING

| Methods | PL | BD | BR | GTF | SV | LV | SH | TC | BC | ST | SBF | RA | HA | SP | HC | CC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **single-scale:** | | | | | | | | | | | | | | | | | |
| RetinaNet-O [25] | 71.43 | 77.64 | 42.12 | 64.65 | 44.53 | 56.79 | 73.31 | 90.84 | 76.02 | 59.96 | 46.95 | 69.24 | 59.65 | 64.52 | 48.06 | 0.83 | 59.16 |
| FR-O [24] | 71.89 | 74.47 | 44.45 | 59.87 | 51.28 | 68.98 | 79.37 | 90.78 | 77.38 | 67.50 | 47.75 | 69.72 | 61.22 | 65.28 | 60.47 | 1.54 | 62.00 |
| RSDet++ [42] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 62.20 |
| Mask R-CNN [27] | 76.84 | 73.51 | 49.90 | 57.80 | 51.31 | 71.34 | 79.75 | 90.46 | 74.21 | 66.07 | 46.21 | 70.61 | 63.07 | 64.46 | 57.81 | 9.42 | 62.67 |
| Oriented-RCNN [17] | 72.15 | **83.02** | 50.18 | 71.27 | 43.33 | 75.56 | 80.78 | **90.87** | 78.34 | 60.18 | 53.80 | 73.36 | 68.47 | 65.70 | 55.33 | 12.31 | 64.66 |
| KLD [45] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 65.18 |
| ReDet [11] | 79.20 | 82.81 | 51.92 | 71.41 | 52.38 | 75.73 | 80.92 | 90.83 | 75.81 | 68.64 | 49.29 | 72.03 | 73.36 | 70.55 | 63.33 | 11.53 | 66.86 |
| Oriented-Rep [13] | 78.95 | 80.17 | 51.68 | 73.90 | **57.36** | 74.72 | 86.43 | 90.86 | 79.33 | **76.86** | 54.51 | **73.76** | 66.83 | **70.89** | 51.62 | 7.71 | 67.22 |
| GSDDet (Ours) | **80.32** | **82.93** | **52.70** | **74.18** | 52.33 | **76.37** | **87.16** | 90.80 | **84.14** | 68.69 | 63.50 | 72.44 | **73.60** | 66.65 | **65.04** | **15.15** | **69.13** |
| **multi-scale:** | | | | | | | | | | | | | | | | | |
| OWSR* [46] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 74.90 |
| ReDet* [11] | **88.51** | **86.45** | **61.23** | 81.20 | **67.60** | 83.65 | **90.00** | 90.86 | 84.30 | 75.33 | **71.49** | 72.06 | 78.32 | 74.73 | 76.10 | 46.98 | 76.80 |
| GSDDet* (Ours) | 87.65 | 86.24 | 60.91 | **81.67** | 67.50 | **84.06** | **90.00** | **90.88** | **86.16** | **77.33** | 71.26 | **76.07** | **78.49** | **75.16** | **76.30** | **53.13** | **77.68** |

TABLE VIII

COMPARISONS WITH STATE-OF-THE-ART METHODS ON HRSC2016 OBB TASK. THE VOC2007 METRICS WERE USED TO EVALUATE AND COMPARE ALL THE METHODS

| Methods | mAP |
|---|---|
| RoI Trans. [7] | 86.20 |
| Gliding Vertex [31] | 88.20 |
| R3Det [32] | 89.26 |
| CSL [29] | 89.62 |
| DAL [47] | 89.77 |
| S2A-Net [8] | 90.17 |
| Oriented RepPoints [13] | 90.38 |
| Oriented R-CNN [17] | 90.40 |
| ReDet [11] | 90.46 |
| GSDDet (Ours) | **90.51** |

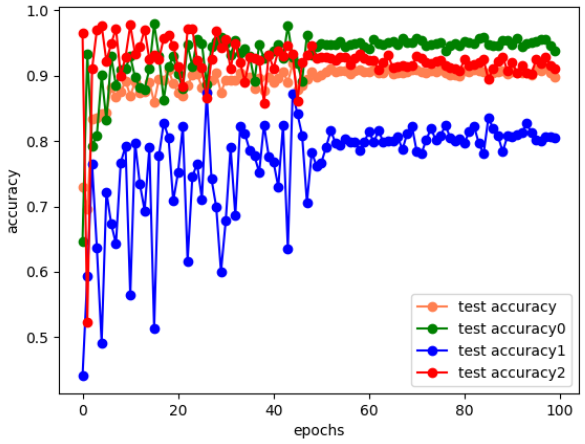



Fig. 7. **The test results of GSD classification.** The graphic depicts the evaluation of accuracy in a coral-based mapping system. The coral is a representation of the overall accuracy, while the different colors signify varying levels of accuracy based on GSD. The green color indicates the accuracy at GSD of 0.3 m, the blue color represents the accuracy at GSD of 0.6 m, and the red color symbolizes the accuracy at GSD of 0.8 m.

Our implementation of the GSDDet technique has allowed us to make significant strides in the complex category, ultimately contributing to an overall increase in mAP when compared with DOTA-v1.0.

*3) Results on HRSC2016:* HRSC2016 contains a lot of thin and long ship instances oriented in arbitrary directions. We compare our method with other state-of-the-art methods in Table VIII. Our method achieved the state-of-the-art performance with **90.51** mAP under the VOC2007 metrics.

### *E. Discussion*

This article introduces a novel approach for object detection in remote sensing images, where the ground sample distance is used as a constraint to reduce confusion between object categories. This constraint is encoded as deep features, which helps compress information across different scales. The effectiveness of this approach has been verified through a series of experiments. In contrast to previous methods that primarily focused on detecting dense objects with rotation against complex backgrounds, our method places greater emphasis on the clues information contained within the remote sensing images themselves.

Although our approach explicitly incorporates ground sample distance constraints into object detection in remote sensing imagery in a direct manner, it is worth noting that previous object detection methods have implicitly relied on ground sample distance information. This can be observed in the data augmentation process, which is particularly evident in the data enhancement, since the data enhancement process is typically accompanied by a change in size of images. In fact, for multiscale training of the DOTA dataset, the initial step involves resizing the images by a factor of 0.5 and 1.5. Our approach builds on this existing knowledge and explicitly incorporates ground sample distance constraints to improve the accuracy of object detection in remote sensing imagery. During the process of changing image size, the data enhancement process not only increases the quantity of data but also modifies the ground sample distance (GSD) of the image. This leads to unintentional inclusion of more GSD information during the training process. This may be the reason why the accuracy improvement of multiscale datasets is lower than that of single-scale datasets when GSD features are incorporated into the network in the same manner. In other words, the addition of GSD information during data augmentation may negatively impact the accuracy of multiscale datasets compared with single-scale datasets. Furthermore, we introduce an attention mechanism to avoid the issue of resolution features being overshadowed by the vast feature network. This attention mechanism helps alleviate the problem to some degree, but it is still an area that requires further investigation. The key challenge is to minimize the loss of both the features while ensuring that the GSD features are not dominated during the process of feature fusion. Hence, it is crucial to strike a balance between the two features for effective feature fusion.

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The data contained within remote sensing images harbor a substantial wealth of information, and the ground sample distance (GSD) remains an underexplored domain with untapped potential. As we progress with our research, our future endeavors will be directed toward delving deeper into the GSD and geographic information inherent in remote sensing images. These aspects hold pivotal importance and warrant further investigation. In essence, the vast reservoir of valuable information concealed within remote sensing images demands our concerted efforts in comprehending and effectively leveraging it to its maximum capacity.

## V. Conclusion

The aim of this article is to develop an object detection method for remote sensing images that reduces incorrect predictions caused by cross-scale confusion. To address this challenge, we introduce a novel ground sample distance (GSD)-guided object detector that leverages GSD constraints as physical references. These constraints are encoded as deep features and seamlessly integrated into a carefully designed object detector architecture. Our method's efficacy is rigorously evaluated through extensive experiments conducted on benchmark datasets, including DOTA-v1.0, DOTA-v1.5, and HRSC2016. The application of this approach improved the performance of both one-stage and two-stage baseline detectors.

## References


[1] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[2] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. CVPR*, 2018, pp. 3974–3983.

[3] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.

[4] Q. He, X. Sun, W. Diao, Z. Yan, D. Yin, and K. Fu, "RETRACTED: Transformer-induced graph reasoning for multimodal semantic segmentation in remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 90–103, Nov. 2022.

[5] J. Wang, W. Yang, H.-C. Li, H. Zhang, and G.-S. Xia, "Learning center probability map for detecting objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4307–4323, May 2021.

[6] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[7] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. CVPR*, 2019, pp. 2849–2858.

[8] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[9] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. 6th Int. Conf. Pattern Recognit. Appl. Methods*, 2017, pp. 324–331.

[10] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[11] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. CVPR*, 2021, pp. 2786–2795.

[12] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[13] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. CVPR*, 2022, pp. 1829–1838.

[14] Z. Guo, X. Zhang, C. Liu, X. Ji, J. Jiao, and Q. Ye, "Convex-hull feature adaptation for oriented and densely packed object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5252–5265, Aug. 2022.

[15] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. CVPR*, 2016, pp. 779–788.

[16] X. Yang, L. Hou, Y. Zhou, W. Wang, and J. Yan, "Dense label encoding for boundary discontinuity free rotation detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 15819–15829.

[17] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 3520–3529.

[18] F. Li, W. P. Kustas, M. C. Anderson, J. H. Prueger, and R. L. Scott, "Effect of remote sensing spatial resolution on interpreting tower-based flux observations," *Remote Sens. Environ.*, vol. 112, no. 2, pp. 337–349, Feb. 2008.

[19] R. R. Hoffman and A. B. Markman, *Interpreting Remote Sensing Imagery: Human Factors*. Boca Raton, FL, USA: CRC Press, 2019.

[20] W. Li, W. Wei, and L. Zhang, "GSDet: Object detection in aerial images based on scale reasoning," *IEEE Trans. Image Process.*, vol. 30, pp. 4599–4609, 2021.

[21] J. Intriligator and P. Cavanagh, "The spatial resolution of visual attention," *Cognit. Psychol.*, vol. 43, no. 3, pp. 171–216, Nov. 2001.

[22] R. Welch, "Spatial resolution requirements for urban studies," *Int. J. Remote Sens.*, vol. 3, no. 2, pp. 139–146, Jan. 1982.

[23] G. J. O. Vermeer, "Factors affecting spatial resolution," *Geophysics*, vol. 64, no. 3, pp. 942–953, May 1999.

[24] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.

[25] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[26] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[27] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[28] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8232–8241.

[29] X. Yang and J. Yan, "Arbitrary-oriented object detection with circular smooth label," in *Proc. ECCV*. Cham, Switzerland: Springer, 2020, pp. 677–694.

[30] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. Asian Conf. Comput. Vis.* Cham, Switzerland: Springer, 2018, pp. 150–165.

[31] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[32] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 4, pp. 3163–3171.

[33] C. Zhu, Y. He, and M. Savvides, "Feature selective anchor-free module for single-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 840–849.

[34] X. Pan et al., "Dynamic refinement network for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11204–11213.

[35] D. A. Quattrochi and M. K. Ridd, "Analysis of vegetation within a semi-arid urban environment using high spatial resolution airborne thermal infrared remote sensing data," *Atmos. Environ.*, vol. 32, no. 1, pp. 19–33, Jan. 1998.

[36] P. M. Atkinson and P. J. Curran, "Choosing an appropriate spatial resolution for remote sensing investigations," *Photogramm. Eng. remote Sens.*, vol. 63, no. 12, pp. 1345–1351, 1997.

[37] S. Benjamin and L. Gaydos, "Spatial resolution requirements for automated cartographic road extraction," *Photogramm. Eng. Remote Sens.*, vol. 56, no. 1, pp. 93–100, 1990.

[38] A. Abdollahi, B. Pradhan, N. Shukla, S. Chakraborty, and A. Alamri, "Deep learning approaches applied to remote sensing datasets for road extraction: A state-of-the-art review," *Remote Sens.*, vol. 12, no. 9, p. 1444, May 2020.


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[39] M. W. Gardner and S. Dorling, "Artificial neural networks (the multilayer perceptron)—A review of applications in the atmospheric sciences," *Atmos. Environ.*, vol. 32, nos. 14–15, pp. 2627–2636, 1998.
[40] C.-Z. Qin, L.-J. Zhan, and A.-X. Zhu, "How to apply the geospatial data abstraction library (GDAL) properly to parallel geospatial raster I/O?" *Trans. GIS*, vol. 18, no. 6, pp. 950–957, Dec. 2014.
[41] Y. Dai, F. Gieseke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3559–3568.
[42] W. Qian, X. Yang, S. Peng, X. Zhang, and J. Yan, "RSDet++: Point-based modulated loss for more accurate rotated object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7869–7879, Nov. 2022.
[43] J. Wang, J. Ding, H. Guo, W. Cheng, T. Pan, and W. Yang, "Mask OBB: A semantic attention-based mask oriented bounding box representation for multi-category object detection in aerial images," *Remote Sens.*, vol. 11, no. 24, p. 2930, Dec. 2019.
[44] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 11830–11841.
[45] X. Yang wtal., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 18381–18394.
[46] C. Li et al., "Learning object-wise semantic representation for detection in remote sensing imagery," in *Proc. CVPR Workshops*, 2019, pp. 20–27.
[47] Q. Ming, Z. Zhou, L. Miao, H. Zhang, and L. Li, "Dynamic anchor learning for arbitrary-oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 3, pp. 2355–2363.
[48] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. ECCV*, 2018, pp. 3–19.
[49] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.
[50] G. Jocher, K. Nishimura, T. Mineeva, and R. Vilariño. (2020). *YOLOV5*. [Online]. Available: https://github.com/ultralytics/yolov5



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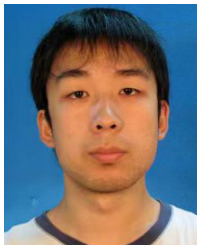
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